

The Specno case study was provided to the Information Systems Department of the University of Cape Town with the consent of the Specno. Specno is an end-to-end digital partner, that validates, s and builds powerful native mobile and web applications to drive business growth and help clients achieve their goals (https://www.specno.com/).

Background & Context

Specno is a small to medium-sized (SME) start-up software development and design agency that services businesses and entrepreneurs. The company has offices in Cape Town, Netherlands and Sweden. Their services include designing applications, web development and enhancing user experience. The company has completed projects globally in industries such as Fintech, eCommerce and Ticketing (Quicket). As a tech start-up, Specno values employee satisfaction and has developed ideas to encourage employee wellness.

The company knows employees have personal and family responsibilities to attend to daily before going to work. Most employees would have to prepare lunch meals while juggling work, personal and family commitments. The company decided to provide meal options to employees in an attempt to improve the working environment. This could improve employee satisfaction, staff morale and engagement at work. Increased employee satisfaction and morale enhance employee efficiency, performance, and productivity. In addressing the issue of meals for employees, Specno decided to provide meals to employees on a daily basis during the week.

Before the pandemic, employees were expected to work and be present in the office. Employees would indicate their availability for attending the meal. This made catering meals for employees easier, as the caterer knew the number of attendees to anticipate. At the beginning of the COVID-19 pandemic, many companies had to enforce policies where workers would work from home. During Covid-19, Specno switched to a hybrid approach to meet safety requirements. This made anticipating meal attendees difficult

for the caterer. Now that the pandemic is over and life has returned to normal, office spaces are beginning to be occupied again. Specno expects employees to alternate days on which they come into the office. This makes it difficult for the caterer to know the number of employees on the premises on any given day.

General problems defined

Before the pandemic, employees would sign up for meals using a spreadsheet. This proved to be a slow and tedious process for both employees and the caterer. They often forget to sign up or forgot they did, thus missing the meal. As a result, the caterer would have to deal with either a shortage or surplus of food. The caterer would also spend an significant amount of time following up on attendance. Specno identified the need to develop a system that will manage the provision of meals to the company's employees.

A hybrid setup has proved to be a challenge for companies that would like to provide their employees with meals when they are in the office. This is because it can be an laborious process to get an accurate number of people that should be catered for. This could also lead to either a shortage or a wastage of food. This created a need for a food management and planning system that will aid the planning and meal provision process in the workplace that will be accurate andmore efficient.

As a Systems Analysis student, you have been tasked with the responsibility of analysing and designing a meal management app to help the caterer plan and schedule meals. In addition, the app should allow employees to book their meals in advance. The objective is to help the company manage its food provision process by providing a platform for the caterers to plan meals and for the employees to book their preferred meals.

Current Process

Currently, making and providing meals to employees begins with the operations manager sending invites to employees via Google Calendar. If the employee will be in the office on the specified day and wants a meal, they accept the invitation. They decline the invite if the employee does not wish to receive a meal. Currently the operations manager compiles a list of employees who accept the invite and forwards it to the caterer via WhatsApp. The caterer then uses this information to count the meals required for that day. This allows the caterer to prepare the meals and the employees would come and eat the meals during lunch time. The current process has

limited flexibility because it does not allow employees to update their availability. This leads to an inaccurate number of people showing up for meals, leading to shortages or a surplus of meals. Shortages of meals may potentially decrease employee satisfaction in the workplace. While surplus results can lead to food wastage, a financial concern for the business. Additionally, the current process makes it difficult for employees to communicate their dietary requirements; specify their availability during lunch time or arrange that food may be set aside to be collected. This is because some employees have meetings during lunchtime and may be unable to collect their meals.

Proposed process

The meal management and booking system will be integrated with the existing Specno system and will have the following functionality:

- The **caterer** must be log on to the system and create a menu (with meal options). Every menu could have more than one meal option. Menus and meals (previously planned) can be stored in a database that can be updated and edited.
- The **caterer** can add meals from the database or plan new meals to compile a menu for a specific day. These menus will be uploaded on a calendar for designated days.
- The **employee** must log in to the system using their work login details.
- **Employees** should be able to book a meal by selecting one of the available meal options from the menu for a specific day.
- **Employees** should be able to select the option of ‘collecting’ their meals if they are not present during lunch time.
- **Employees** should be able to specify dietary restrictions such as allergies, preferences, and intolerances. This information can help the caterer to plan the meals accordingly and avoid any potential allergic reactions.
- **Employees** should be able to make bookings for visiting guests coming to the company. In such a case they should be able collect more than one meal option from the specific menur for the day.
- The system should send notification reminders about meal bookings with a link that directs employees to the web page where they select the days they will be at work and book the meals they will be attending. (time for reminders – end of every week – Friday 10am)

- There should be a cut-off date and time for making meal bookings for a specific day to allow the caterer to plan and prepare meals.
- The system should allow employees to **provide** feedback on the meals provided. This feedback could help the caterer to improve the meal options and adjust as needed.
- Management should be able to generate reports for insights into the process and make informed decisions when planning and preparing budgets.

In summary, the caterer sets plans for the meal days and posts menus on a weekly basis. The employees will access the set meal days and sign up for meals on preferred days. Once sign-ups close, the caterer compiles a list of people per meal choice. When the meals are ready, the employees can come and have their meals. Meals will be set aside for collection, for those employees who indicated that they will be unavailable during lunchtime.

This system can potentially be used by other similar sized companies as there is a shared need across the start-up ecosystem.

Benefits of the proposed systems

The proposed business process makes booking meals more efficient by introducing a meal booking system to eliminate the need for the operations manager to communicate information between the caterer and the employees. In the new process, the operations manager will have more time to focus on other work responsibilities. The meal booking system will provide a more accurate picture of the number of people present for meals on a particular day, thereby reducing food wastage and saving the business money. By allowing employees to have a meal set aside for them when they have other engagements during lunchtime, the business values their time and preferences, which can help boost morale and instil trust in the company. The proposed business process is more accommodating.

With this system, Specno will be able to provide meals to the employees while providing the caterer with the best environment to carry this process out. This will help them keep their business operations running more efficiently.

Proposed solution

The focus for this project will be the development of a meal management system (creating of menus, as well as the scheduling and booking of meals from these menus). It will be developed in-house by Specno's IT team. This will be done as a two phase

approach where a beta version will be released for testing by employees for a month. The final improved phase that will incoporate the user comments and feedback will then be implemented. You can assume that Specno already has an inventory management system that will ensure that ingredients for the meals can be replenished. Specno also have an up and running HR system.

Develop a meal management system that automates the process of planning, scheduling, and providing meals